

# Universität Freiburg – Institut für Physik

Applied Theoretical Physics

Prof. Dr. Michael Moseler

Freiburg, 01.06.2026

**Due: 08.06.2026**

## Computational Physics: Materials Science - Exercise 5, SS 2026

Email: andreas.doell@physik.uni-freiburg.de, severin.sylla@iwm.fraunhofer.de

Tutorials: Andreas Doell, Severin Sylla

---

In this exercise, we extend the MD simulation from Exercise 4 to estimate the heat capacity of sodium. We calculate the isochoric heat capacity  $C_V$  from MD simulations using the following formulas:

### Method 1 — Energy fluctuations

$$C_V(T) = \frac{\langle \delta E_{\text{tot}}^2 \rangle}{k_B T^2} = \frac{\langle E_{\text{tot}}^2 \rangle - \langle E_{\text{tot}} \rangle^2}{k_B T^2} \quad (1)$$

### Method 2 — Finite differences

$$C_V(T) \approx \frac{\langle E_{\text{tot}}(T + \Delta T) \rangle - \langle E_{\text{tot}}(T - \Delta T) \rangle}{2\Delta T} \quad (2)$$

**A few tips:** You might find it useful to work with the Atomic Simulation Environment (ASE, <https://ase-lib.org/ase/atoms.html>) and use modules such as `ase.Atoms`, `ase.build` and `ase.md.langevin` to keep your code reliable and comprehensible. Since the calculations required for this exercise can take a lot of computation time, it makes sense to start with a small number of simulations to develop your code, and then run a larger number of simulations to study the physics of the system. However, be aware that at low temperatures, it takes more time until the system reaches a stationary state than at high temperatures. You can play with the number of simulated timesteps and the thermostat's "aggressiveness" to obtain reasonable results with limited computational power and time. For more accurate results, consider running an equilibration period, followed by a production run with large number of steps. If your simulations are running smoothly you can even increase the number of particles which will help with convergence.

### Problem 5. NVT ensemble and isochoric heat capacity

As a starting point, use your working NVT simulation of  $N = 27$  Na atoms with the Berendsen thermostat and Morse potential with cutoff from exercise sheet 4. As in sheet 4, use **numba** to speed up your computations. use a timestep of  $\Delta t = 1\text{fs}$ .

- (a) Run the NVT simulation with a Berendsen thermostat and with a Langevin thermostat. Analyze the result. Are the thermostats working correctly for this system? What differences do you expect between the two simulations? Compare the results of your simulations.
- (b) Calculate the heat capacity  $C_V$  of the system with method 1 (energy fluctuations) at temperature  $T = 5000\text{K}$ . Compare both Langevin and Berendsen NVT dynamics. Compare your results to the ideal gas law.

- (c) Run Langevin NVT simulations at a few different temperatures, covering a range from below room temperature to the 5000K used in (b). For each simulation, calculate  $C_V(T)$  with method 1. Examine the trajectories in ovito and plot the total energy as a function of time.
- (d) Compare the results for  $C_V(T)$  to an ideal gas and to the Dulong-Petit law. You might notice that our simulations return unexpected results in some cases. Can you explain why our simulation method cannot describe these situations well?
- (e) Repeat the simulation with a Berendsen thermostat and compare to the results obtained with the Langevin thermostat.
- (f) Calculate  $C_V$  for  $T = 300\text{K}$  and  $5000\text{K}$  using method 2 and each thermostat. Discuss the results. How do both methods compare?

### What to submit

Hand in a zip file on ILIAS containing the following files:

- your Python source code
- a short report in PDF format with your main results

A good submission contains:

- A detailed description of your code and the goal of the exercise.
- Make sure that you include the plots for validation and explain why it works and how it works.
- Finish up your submission with illustrative ovito plots of your cluster at different temperatures which can demonstrate the melting.

### How to prepare for the exercise group

- Make sure you can explain your results and the code.
- Prepare some exemplary trajectories that you can show in ovito.
- Keep your chat history with an AI chatbot to explain in the tutorial how you **interacted** with it to develop your code.